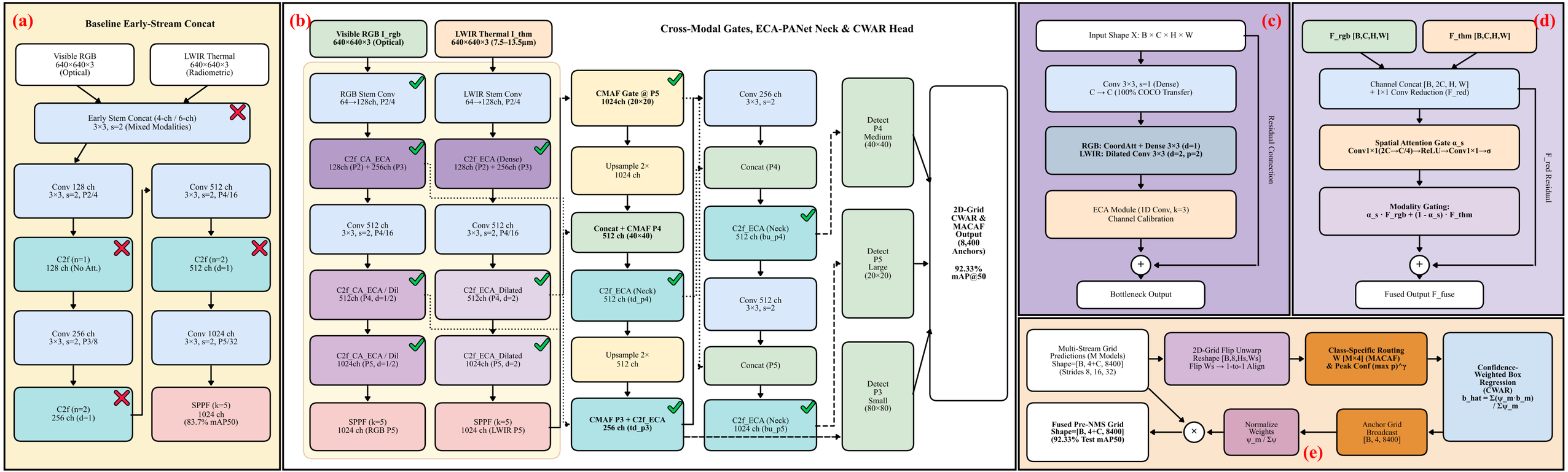

(a)
Baseline Early-Stream Concat
Visible RGB 640×640×3 (Optical)
LWIR Thermal 640×640×3 (Radiometric)
Early Stem Concat (4-ch / 6-ch) 3×3, s=2 (Mixed Modalities)
Conv 128 ch 3×3, s=2, P2/4
Conv 512 ch 3×3, s=2, P4/16
C2f (n=1) 128 ch (No Att.)
C2f (n=2) 512 ch (d=1)
Conv 256 ch 3×3, s=2, P3/8
Conv 1024 ch 3×3, s=2, P5/32
C2f (n=2) 256 ch (d=1)
SPPF (k=5) 1024 ch (83.7% mAP50)
(b)
Cross-Modal Gates, ECA-PANet Neck & CWAR Head
Visible RGB I_rgb 640×640×3 (Optical)
LWIR Thermal I_thm 640×640×3 (7.5–13.5µm)
RGB Stem Conv 64→128ch, P2/4
LWIR Stem Conv 64→128ch, P2/4
C2f_CA_ECA 128ch (P2) + 256ch (P3)
C2f_ECA (Dense) 128ch (P2) + 256ch (P3)
Conv 512 ch 3×3, s=2, P4/16
Conv 512 ch 3×3, s=2, P4/16
C2f_CA_ECA / Dil 512ch (P4, d=1/2)
C2f_ECA_Dilated 512ch (P4, d=2)
C2f_CA_ECA / Dil 1024ch (P5, d=1/2)
C2f_ECA_Dilated 1024ch (P5, d=2)
SPPF (k=5) 1024 ch (RGB P5)
SPPF (k=5) 1024 ch (LWIR P5)
CMAF Gate @ P5 1024ch (20×20)
Upsample 2× 1024 ch
Concat + CMAF P4 512 ch (40×40)
C2f_ECA (Neck) 512 ch (td_p4)
Upsample 2× 512 ch
CMAF P3 + C2f_ECA 256 ch (td_p3)
Conv 256 ch 3×3, s=2
Concat (P4)
C2f_ECA (Neck) 512 ch (bu_p4)
Conv 512 ch 3×3, s=2
Concat (P5)
C2f_ECA (Neck) 1024 ch (bu_p5)
Detect P4 Medium (40×40)
Detect P5 Large (20×20)
Detect P3 Small (80×80)
2D-Grid CWAR & MACAF Output (8,400 Anchors)
92.33% mAP@50
(c)
Input Shape X: B × C × H × W
Conv 3×3, s=1 (Dense) C → C (100% COCO Transfer)
RGB: CoordAtt + Dense 3×3 (d=1) LWIR: Dilated Conv 3×3 (d=2, p=2)
ECA Module (1D Conv, k=3) Channel Calibration
Bottleneck Output
Residual Connection
(d)
F_rgb [B,C,H,W]
F_thm [B,C,H,W]
Channel Concat [B, 2C, H, W] + 1×1 Conv Reduction (F_red)
Spatial Attention Gate α_s Conv1×1(2C→C/4)→ReLU→Conv1×1→σ
Modality Gating: α_s · F_rgb + (1 - α_s) · F_thm
Fused Output F_fuse
F_red Residual
(e)
Multi-Stream Grid Predictions (M Models) Shape=[B, 4+C, 8400] (Strides 8, 16, 32)
2D-Grid Flip Unwarp Reshape [B,8,Hs,Ws] Flip Ws → 1-to-1 Align
Class-Specific Routing W [M×4] (MACAF) & Peak Conf (max p)^γ
Confidence-Weighted Box Regression (CWAR) b_hat = Σ(ψ_m·b_m) / Σψ_m
Fused Pre-NMS Grid Shape=[B, 4+C, 8400] (92.33% Test mAP50)
×
Normalize Weights ψ_m / Σψ
Anchor Grid Broadcast [B, 4, 8400]